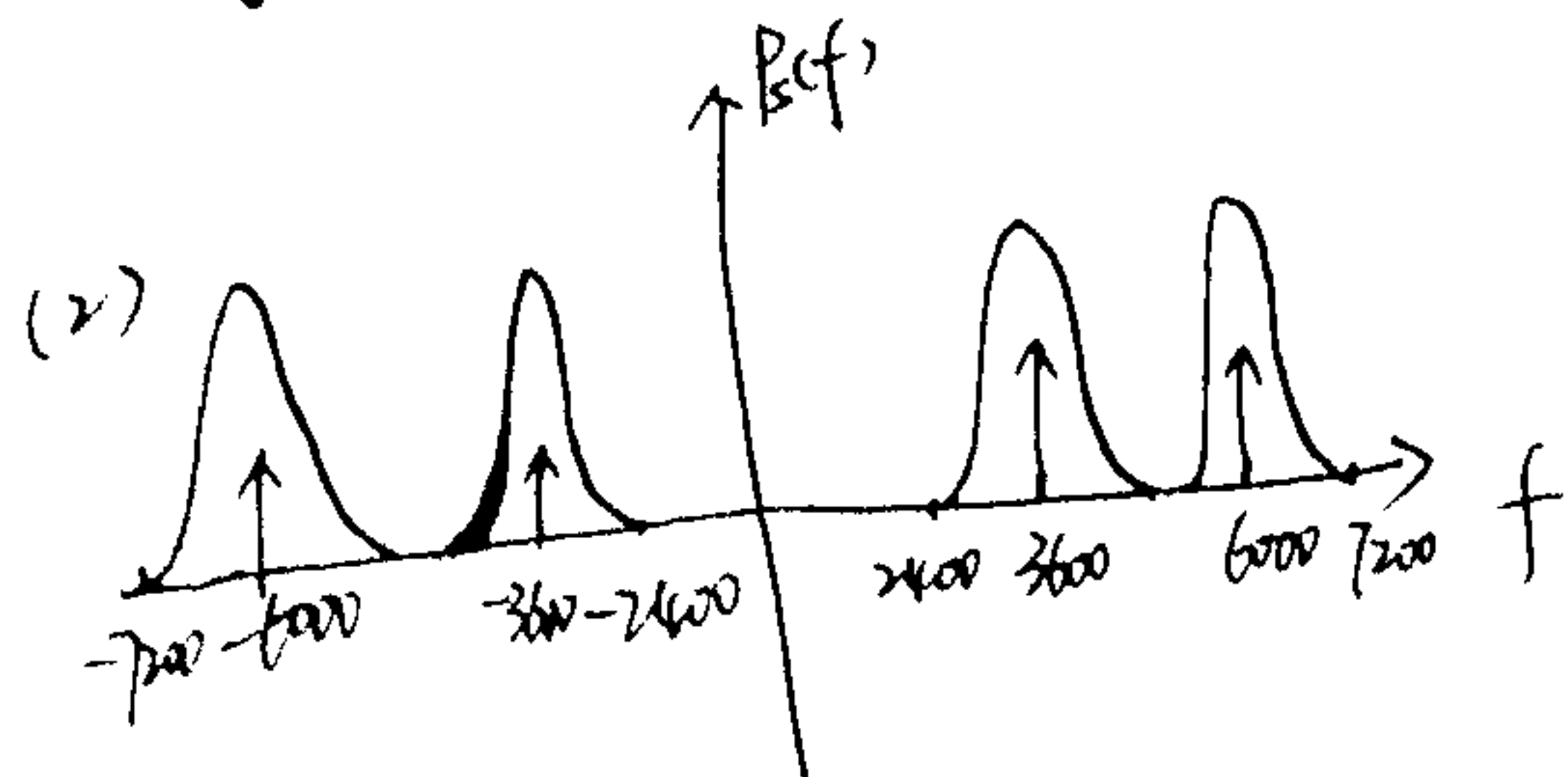
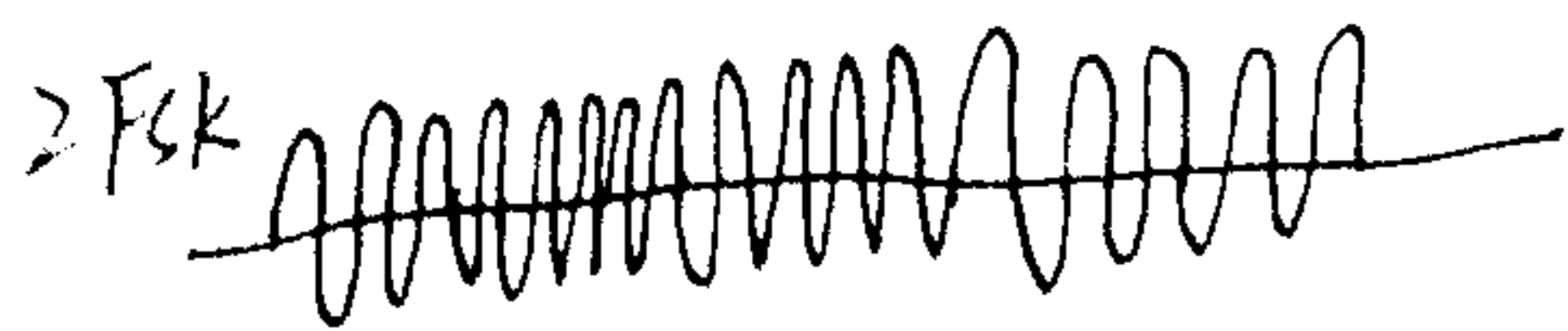
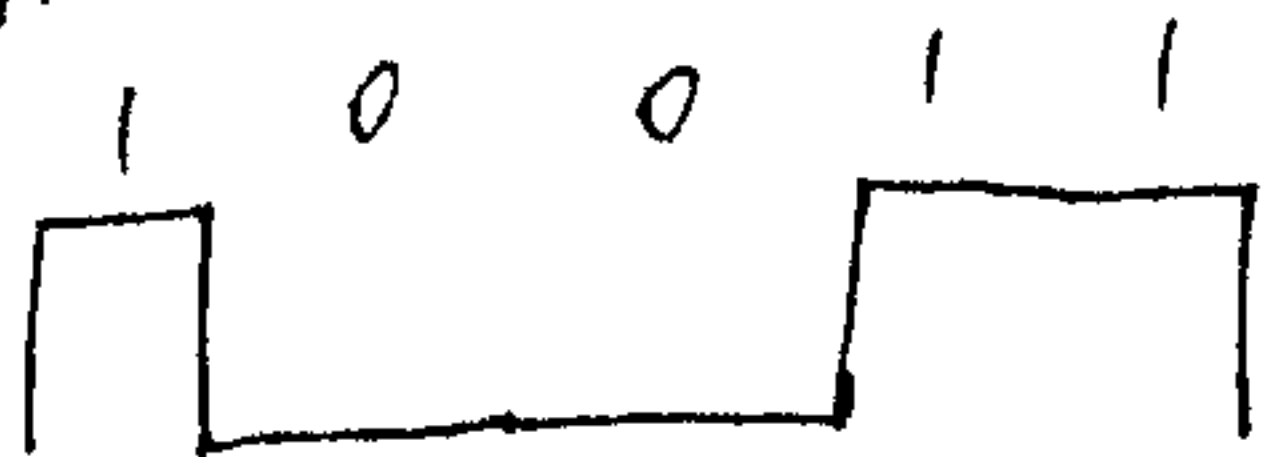


6-2.

(1) $f_1 = 3600$ $f_2 = 6000$ $R_s = 1200$

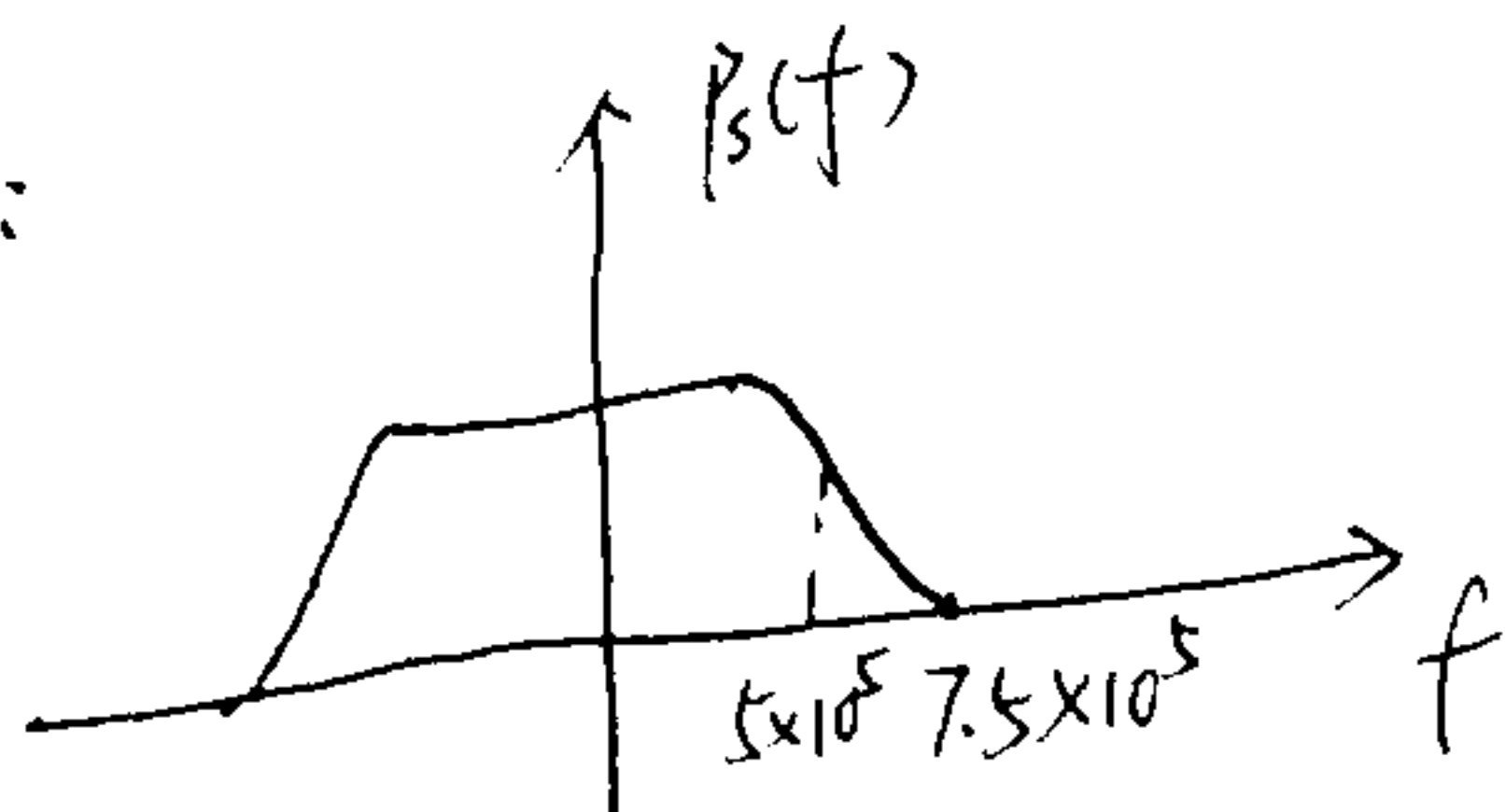


(3). $B_{2FSK} = |f_1 - f_2| + 2B_b = 6000 - 3600 + 2400 = 4800 \text{ Hz}$

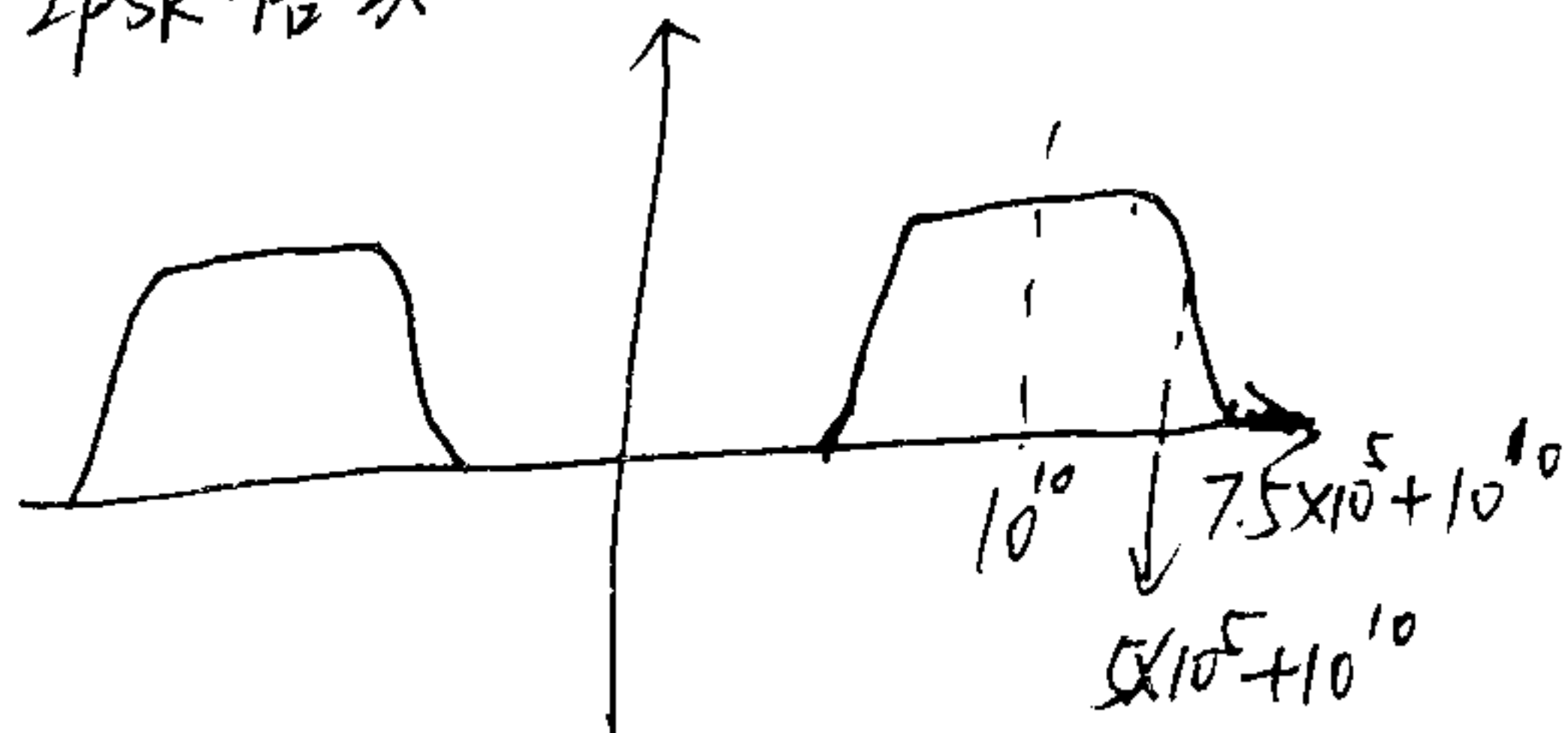
6-3. $R_b = (1+2) \cdot \frac{R_s}{2} = 7.5 \times 10^5 \text{ bit/s/Hz}$

$B_{2PSK} = 2R_b = 1.5 \times 10^6 \text{ Hz}$

基带信号:



2PSK 信号:



6-5(1) 双比特码元与载波的对应关系: (B方式)

00 $\rightarrow -\frac{3}{4}\pi$

10 $\rightarrow -\frac{\pi}{4}$

11 $\rightarrow \frac{\pi}{4}$

01 $\rightarrow \frac{3}{4}\pi$

(2) $R_s = \frac{1}{T_s} = \frac{1}{2T_b} = \frac{R_b}{2}$

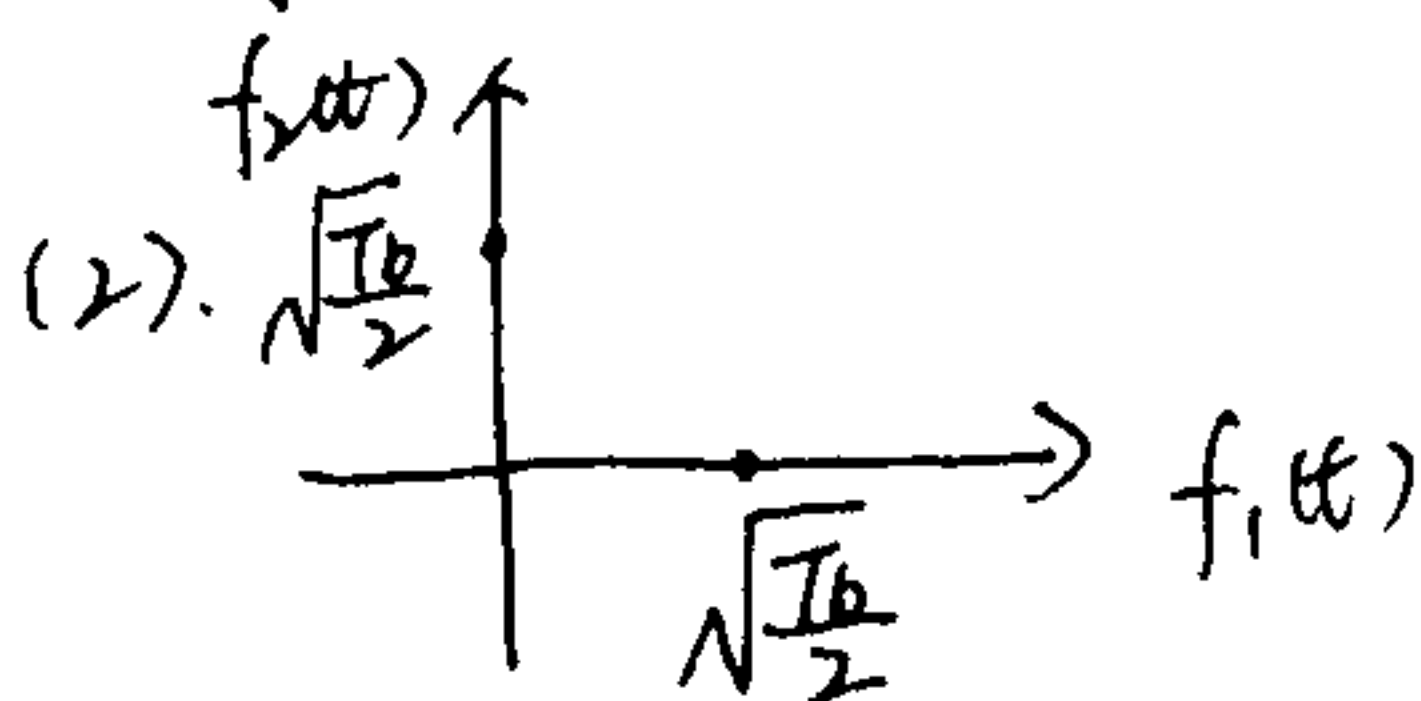
$\therefore B = 2R_s = R_b$

$\therefore \eta = \frac{R_b}{B} = \frac{R_b}{R_b} = 1$

6-10.

(1) $f_1(t) = \sqrt{\frac{2}{T_b}} s_1(t)$

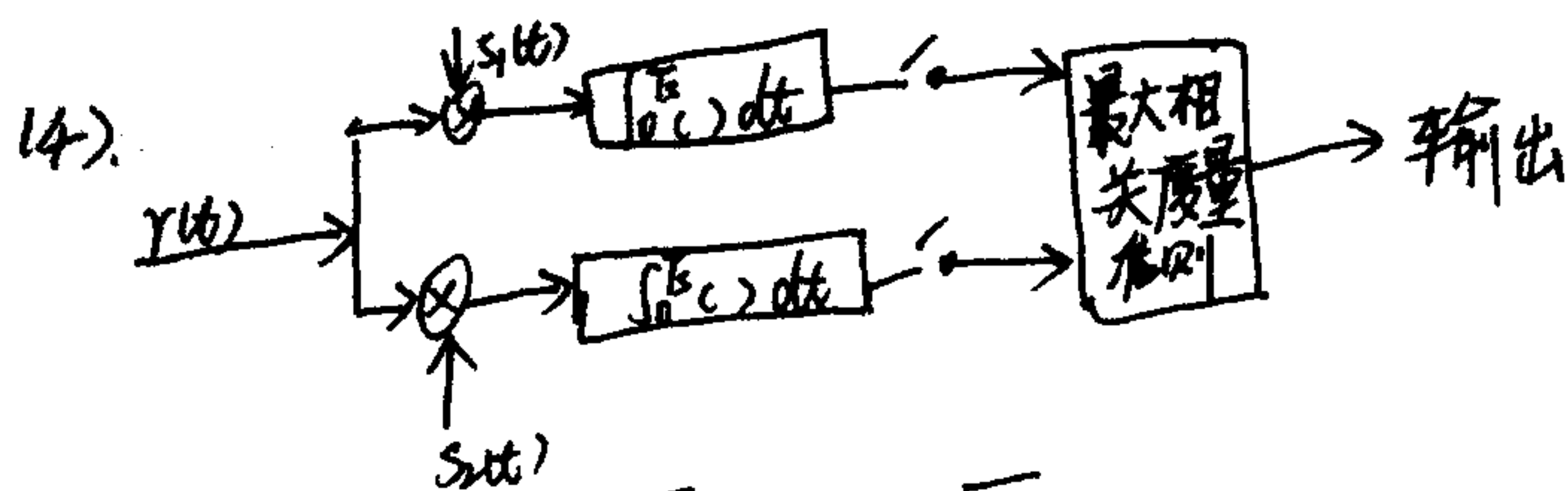
$f_2(t) = \sqrt{\frac{2}{T_b}} s_2(t)$



欧氏距离 $d = \sqrt{2} \cdot \sqrt{\frac{T_b}{2}} = \sqrt{T_b}$

①

(3) $E_1 = \frac{T_b}{2}$, $E_2 = \frac{T_b}{2}$, $E_b = E_1 + E_2 = \frac{T_b}{2}$



(5) $P(e|s_1) = Q(\sqrt{\frac{E_b}{N_0}}) = Q(\sqrt{\frac{T_b}{2N_0}})$

(6) $P_b = P(s_1)P(e|s_1) + P(s_2)P(e|s_2) = Q(\sqrt{\frac{E_b}{N_0}}) = Q(\sqrt{\frac{T_b}{2N_0}})$

6-14.

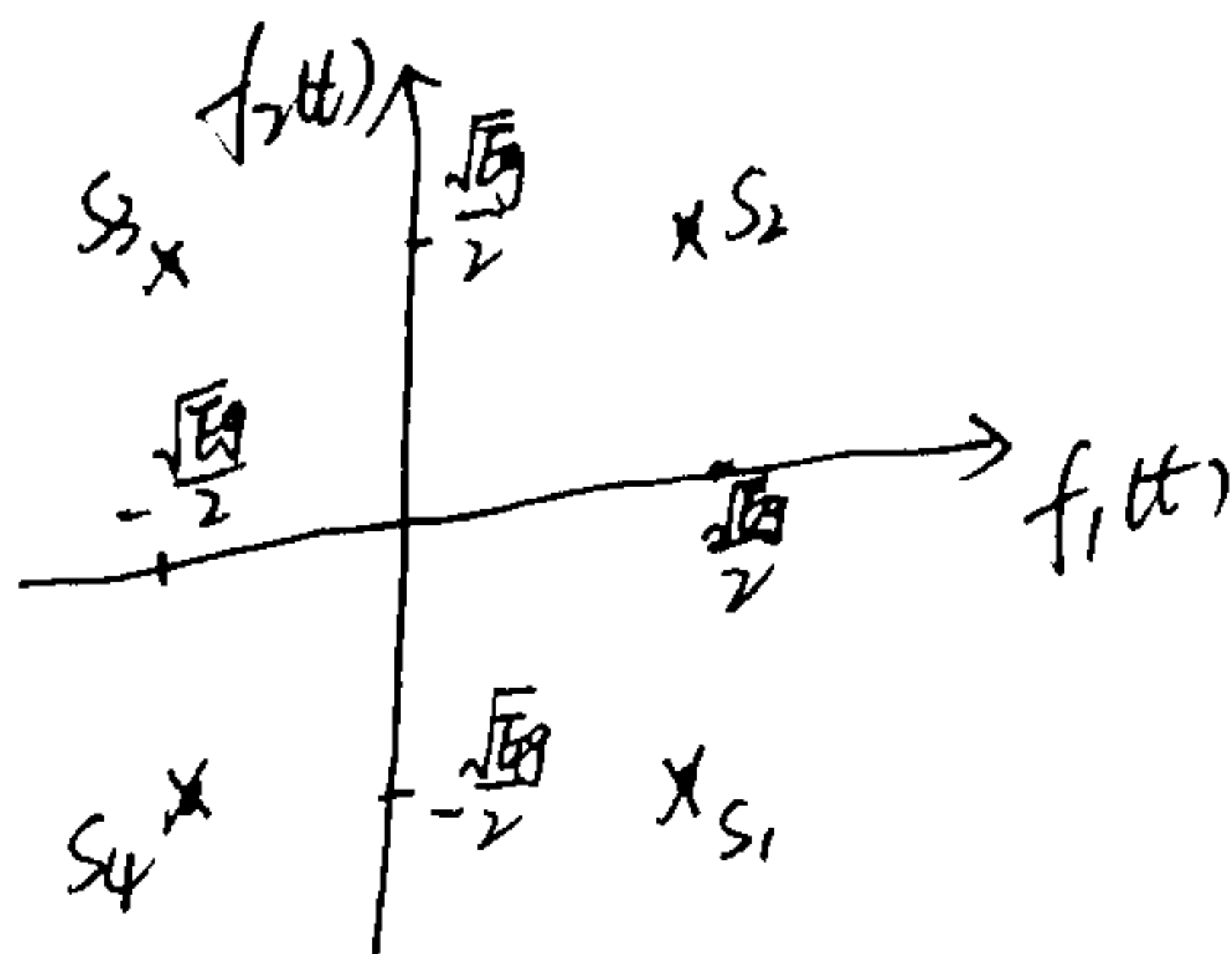
1) $s_1(t) = g_T(t) [\frac{\sqrt{2}}{2} \cos 2\pi f_c t + \frac{\sqrt{2}}{2} \sin 2\pi f_c t]$

$= \frac{\sqrt{E_b}}{2} f_1(t) - \frac{\sqrt{E_b}}{2} f_2(t)$

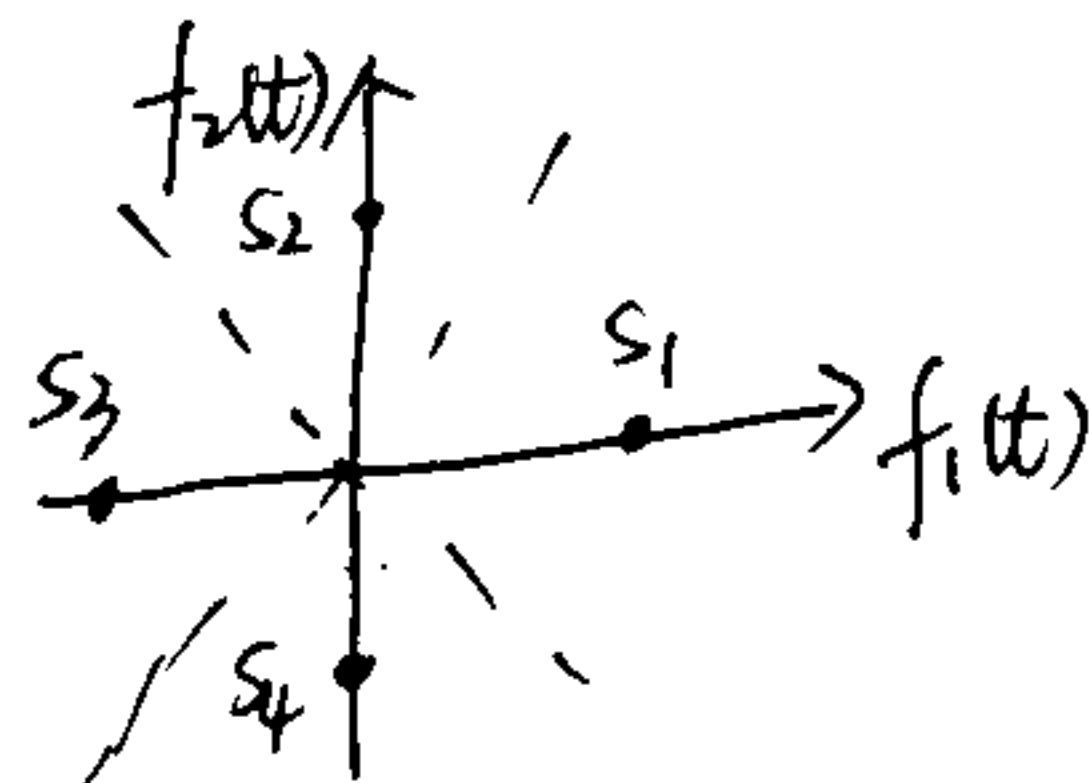
$s_2(t) = \frac{\sqrt{E_b}}{2} f_1(t) + \frac{\sqrt{E_b}}{2} f_2(t)$

$s_3(t) = -\frac{\sqrt{E_b}}{2} f_1(t) + \frac{\sqrt{E_b}}{2} f_2(t)$

$s_4(t) = -\frac{\sqrt{E_b}}{2} f_1(t) - \frac{\sqrt{E_b}}{2} f_2(t)$



(2) $s_1(t) = \sqrt{\frac{E_b}{2}} f_1(t)$ $s_2(t) = \sqrt{\frac{E_b}{2}} f_2(t)$
 $s_3(t) = -\sqrt{\frac{E_b}{2}} f_1(t)$ $s_4(t) = -\sqrt{\frac{E_b}{2}} f_2(t)$



(3) 当矢量图为(1)时, 设 $\vec{r} = (r_1, r_2)$ 为接收矢量,

当 $r_1 > 0, r_2 > 0$, 判决为 s_2 ; 当 $r_1 > 0, r_2 < 0$, 判决为 s_1 ;

当 $r_1 < 0, r_2 > 0$, 判决为 s_3 ; 当 $r_1 < 0, r_2 < 0$, 判决为 s_4 .

当矢量图为(2)时, 设 $\theta = \arctan \frac{r_2}{r_1}$

当 $-\frac{\pi}{4} < \arctan \theta < \frac{\pi}{4}$ 时, 判决为 s_1 ; 当 $\frac{\pi}{4} < \arctan \theta < \frac{3\pi}{4}$ 时, 判决为 s_2 ;

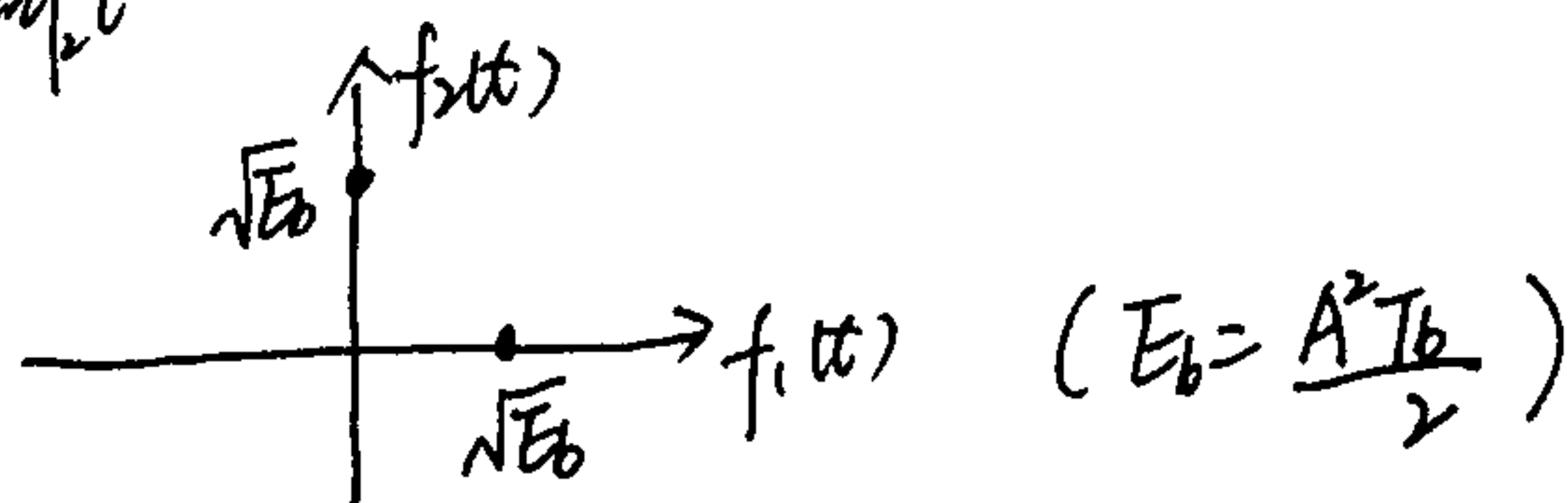
当 $\frac{3\pi}{4} < \arctan \theta < \frac{5\pi}{4}$ 时, 判决为 s_3 ; 当 $\frac{5\pi}{4} < \arctan \theta < \frac{7\pi}{4}$ 时, 判决为 s_4

6-16.

(1) $f_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \frac{A \cos 2\pi f_c t}{\sqrt{\frac{A^2 T_b}{2}}} = \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t$

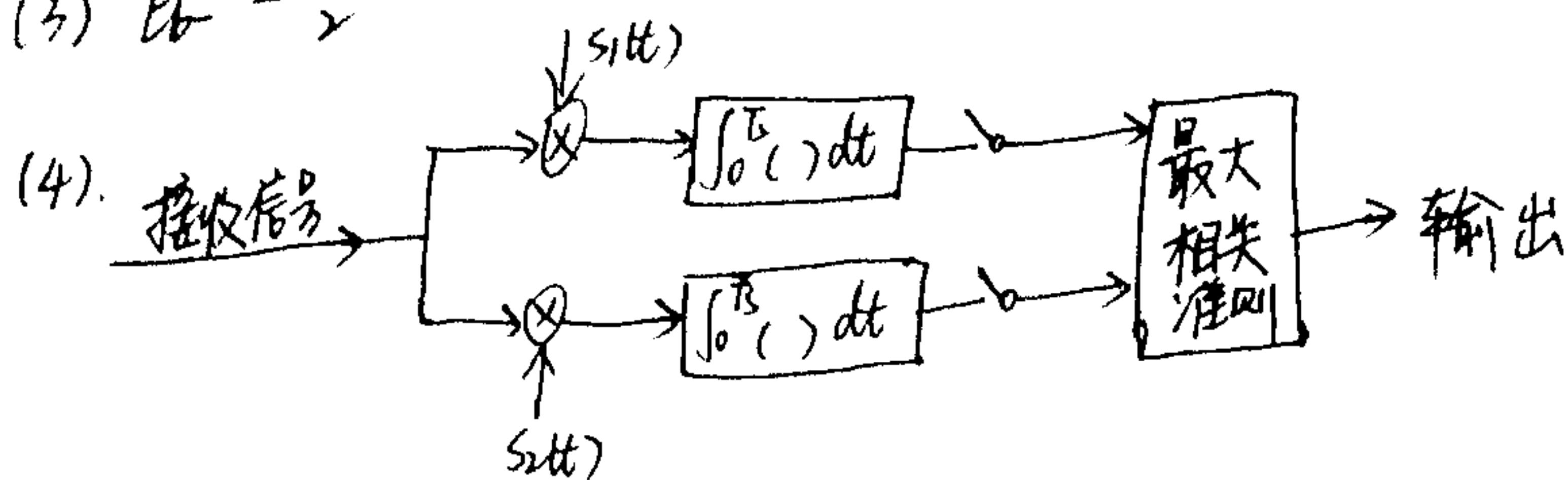
$f_2(t) = \frac{s_2(t)}{\sqrt{E_2}} = \sqrt{\frac{2}{T_b}} \cos 2\pi f_c t$

信号星座图:



$$(2) d_{12} = \sqrt{2E_b} = \sqrt{A^2 T_b}$$

$$(3) E_b = \frac{A^2 T_b}{2}$$



$$P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right) = Q\left(\sqrt{\frac{A^2 T_b}{2N_0}}\right)$$

6-19.

$$(1) \because P_b \leq 10^{-5} \quad (2) Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \leq 10^{-5}$$

$$\therefore \sqrt{\frac{2E_b}{N_0}} \geq 4.265 \Rightarrow E_b \geq 18.1 \times \frac{N_0}{2} = 1.82 \times 10^{-9}$$

$$\text{又} \because E_b = \frac{1}{2} a^2 T_b = \frac{1}{2} a^2 \cdot \frac{1}{P_b} \geq 1.82 \times 10^{-9}$$

$$\therefore a \geq 0.06$$

$$(2) \text{接收功率: } E_b P_b = 1.82 \times 10^{-9} \times 10^4 = 1.82 \times 10^{-5}$$

$$\text{发送功率: } 1.82 \times 10^{-5} \times 10^6 = 18.20 \text{ W}$$

6-20. 解: 设最多传 L km,

$$(1) 2ASK: P_b = Q\left(\sqrt{\frac{r}{2}}\right) \leq 10^{-5} \Rightarrow \sqrt{\frac{r}{2}} \geq 4.265 \Rightarrow r \geq 36.38$$

$$\text{又} \because r = \frac{a^2}{2N_0 B_{\text{BPF}}} \quad \text{且} \quad \frac{a^2}{2} = \frac{A^2}{2} \cdot 10^{-\frac{L}{10}}$$

$$\therefore 50 \cdot 10^{-\frac{L}{10}} \cdot \frac{1}{10^{-8} \times 4 \times 10^3} \geq 36.38$$

$$\therefore L = 45.36 \text{ km}$$

$$(2) 2PSK: Q(\sqrt{r}) \leq 10^{-5} \Rightarrow r \geq 9.1$$

$$r = \frac{A^2}{2} \cdot 10^{-\frac{L}{10}} / N_0 B \geq 9.1$$

$$\therefore 50 \cdot 10^{-\frac{L}{10}} \geq 9.1 \times 10^{-8} \times 4 \times 10^3$$

$$\therefore L \leq 51.38 \text{ km}$$

$$(3) \because |f_1 - f_2| + 2B_{\text{基}} = 4 \times 10^3 \quad \text{且} \quad |f_1 - f_2| = 2B_{\text{基}}$$

$$\therefore B_{\text{基}} = 1 \times 10^3$$

$$\therefore B_{\text{BPF}} = 2B_{\text{基}} = 2 \times 10^3$$

$$\text{又} \because P_b = Q(\sqrt{r}) \leq 10^{-5} \Rightarrow r \geq 18.19$$

$$\therefore \frac{A^2}{2} \cdot 10^{-\frac{L}{10}} / N_0 B_{\text{BPF}} \geq 18.19 \Rightarrow L_{\text{max}} = 51.38 \text{ km}$$

(3)

6-21. 发端信号功率为: $\frac{A^2}{2} = \frac{16 \times 10^{-4}}{2} = 8 \times 10^{-4} \text{ (W)}$

收端接收功率为: $8 \times 10^{-4} \times 10^{-6} = 8 \times 10^{-10} \text{ (W)}$

$$P_b = \frac{1}{2} e^{-\frac{r}{4}} = \frac{1}{2} e^{-\frac{8 \times 10^{-10}}{4 \times 6 \times 10^{-10} \times 4 \times 10^6}} = \frac{1}{2} e^{-\frac{25}{5}} = 1.2 \times 10^{-4}$$

6-22.

(1) $B = (1+\alpha) \cdot \frac{R_s}{2} = (1+0.25) \cdot \frac{10^6}{2} = 6.25 \times 10^5 \text{ Hz}$

(2) 2PSK: $B_{2PSK} = 2B_{基} = 1.25 \times 10^6 \text{ Hz}$

16ASK: $R_s = \frac{R_b}{\log_2 16} = 2.5 \times 10^5 \text{ Baud}$

$$B' = (1+\alpha) \cdot \frac{R_s}{2} = 1.56 \times 10^5 \text{ Hz}$$

$$\therefore B_{16ASK} = 2B' = 3.12 \times 10^5 \text{ Hz}$$

16PSK: $B_{16PSK} = 3.12 \times 10^5 \text{ Hz}$

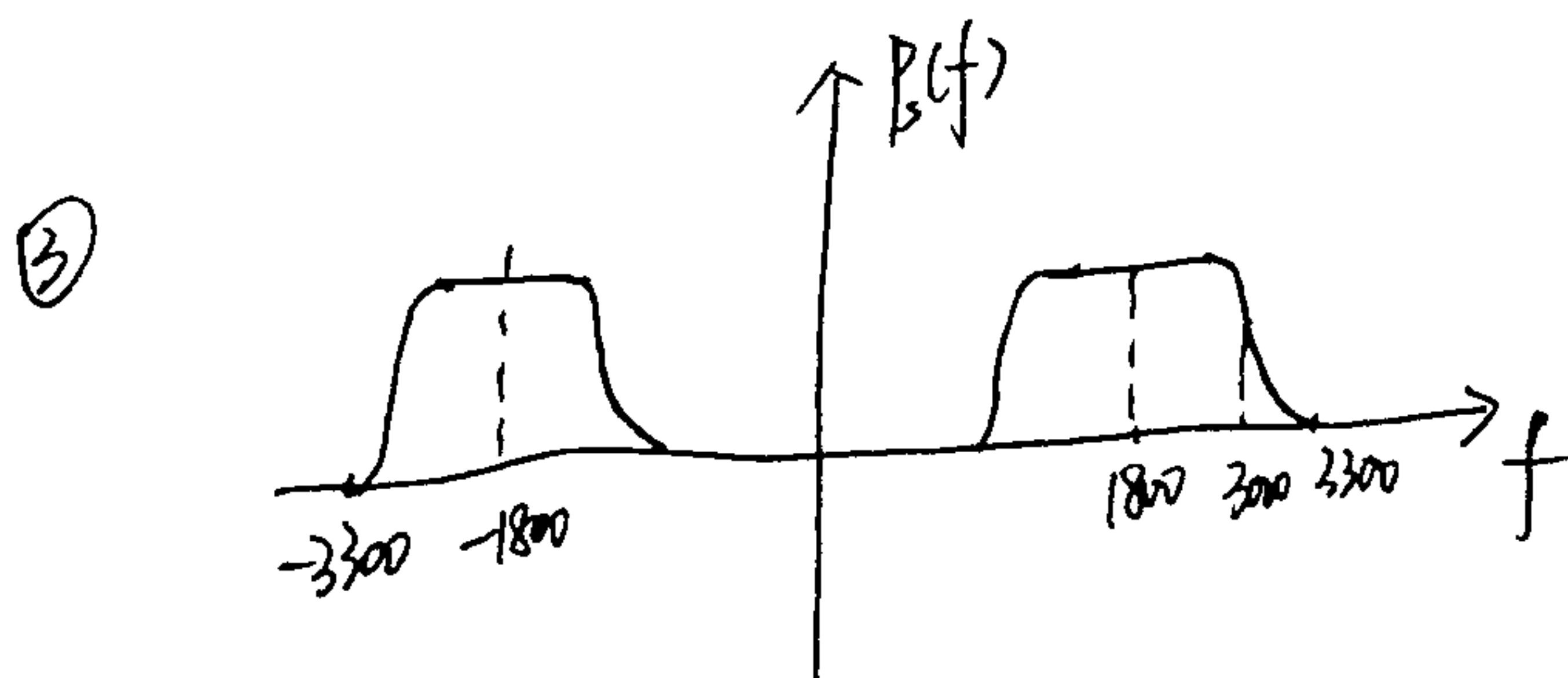
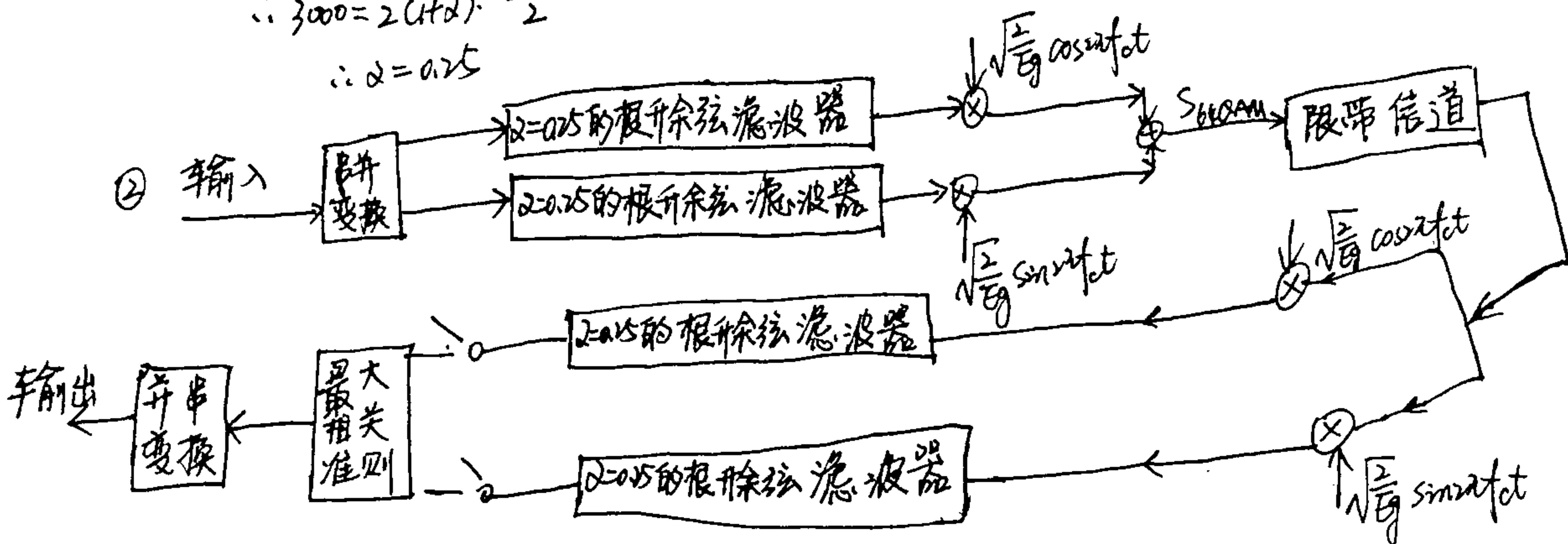
16QAM: $B_{16QAM} = 3.12 \times 10^5 \text{ Hz}$

6-29. ① $R_b = R_s \cdot \log_2 M$
 $\therefore 14400 = 2400 \log_2 M \Rightarrow M = 64$
 $f_c = \frac{3300 + 300}{2} = 1800 \text{ Hz}$

$\therefore B = 2(1+\alpha) \cdot \frac{R_s}{2}$

$\therefore 3000 = 2(1+\alpha) \cdot \frac{2400}{2}$

$\therefore \alpha = 0.25$



④

6-31

$$(1) f_m = 4 \times 10^6 \text{ Hz}$$

$$f_s = 2f_m = 8 \times 10^6 \text{ Hz}$$

$$R_b = 8 \times 10^6 \times 8 = 64 \times 10^6 \text{ bit/s}$$

$$R_s = R_b = 6.4 \times 10^7 \text{ Baud}$$

$$B = 2(1+\alpha) \cdot \frac{R_s}{2} = 7.68 \times 10^7 \text{ Hz}$$

$$\eta = \frac{R_b}{B} = \frac{6.4 \times 10^7}{7.68 \times 10^7} = 0.833 \text{ bit/s/Hz}$$

$$(2) R_b = 6.4 \times 10^7 \text{ bit/s}$$

$$R_s = \frac{R_b}{\log_2 M} = 8 \times 10^6 \text{ Baud}$$

$$B = 2(1+\alpha) \cdot \frac{R_s}{2} = 9.6 \times 10^6 \text{ Hz}$$

$$\eta = \frac{R_b}{B} = \frac{6.4 \times 10^7}{9.6 \times 10^6} = \frac{20}{3} = 6.67 \text{ bit/s/Hz}$$